Water Reminder

Interactive 3D Fluid Hydration Tracker & Lock-Screen Alarm Companion

A dark-mode glassmorphic Android application combining real-time dual-wave fluid physics, personalized intake calibration, 7-day habit analytics, and full-screen wake-up alar

ROLE
UI/UX & Android Product Designer

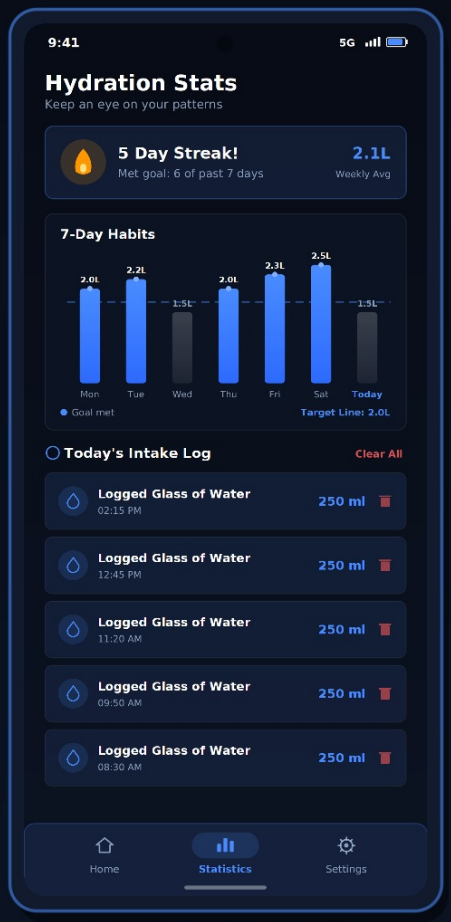
PLATFORM
Android Native (Jetpack Compose + M3

AESTHETIC
Luxury Dark Sapphire Glassmorphism

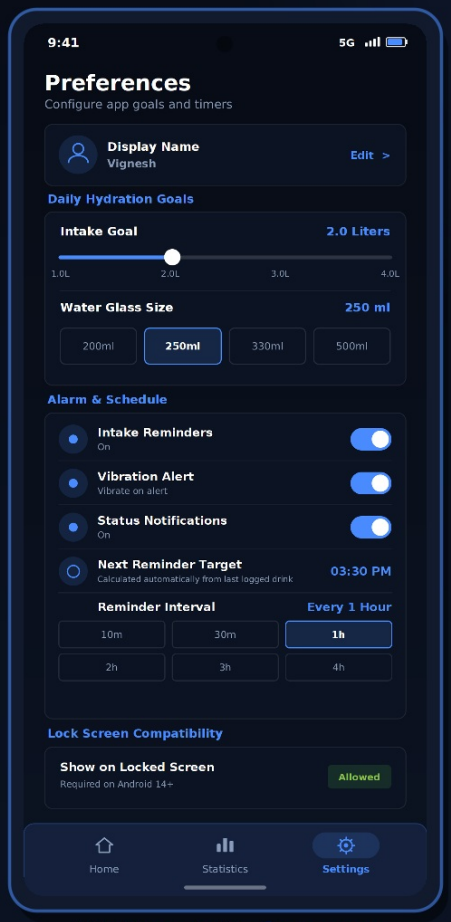
PERSISTENCE
Room SQLite v4 + Exact AlarmManager



01. Home Dashboard



02. Hydration Stats



03. Smart Preferences



04. Lock-Screen Alert

UX Case Study: Solving Hydration Drop-Off

Why standard water tracking apps fail—and how tactile visual feedback builds lasting daily habits.

THE PROBLEM STATEMENT

Passive Notification Fatigue & Boring Data Entry

1. Ignored Status-Bar Alerts:
Standard silent or banner notifications blend into dozens of daily social/work messages and are swiped away without action.

2. Sterile Utility Interfaces:
Flat progress bars and spreadsheet-style forms make logging water feel like a chore rather than a rewarding self-care ritual.

3. Rigid Scheduling:
Fixed timers fail to adapt when a user drinks water early, causing redundant back-to-back notifications.

THE DESIGN SOLUTION

Tactile 3D Fluid Physics + Lock-Screen Wake-Up UX

1. Interactive 3D Bottle Canvas:
Custom dual-wave sine curve physics, rising air bubbles, and real-time liter markings provide immediate visual gratification.

2. Google-Clock Inspired Lock-Screen Alert:
Full-screen wake-up activity overlays the lock screen with looping audio, custom haptics, and 1-tap logging or 10m snooze.

3. Dynamic Next-Drink Recalculation:
Every logged drink automatically recalculates and schedules the next exact alarm based on the user's chosen interval.

END-TO-END USER FLOW ARCHITECTURE

STEP 1: CALIBRATE

Set Daily Goal (1.0L–4.0L), Glass Size (200–500ml) & Reminder Interval (10m–4h)

→

STEP 2: TRACK & LOG

Tap glowing FAB on Home to log +250ml; watch 3D fluid level rise smoothly

→

STEP 3: WAKE-UP ALERT

Full-screen alarm triggers with custom tone & vibration; Log or Snooze in 1 tap

→

STEP 4: ANALYZE HABITS

Review 7-Day Bar Chart, consecutive Streak count & timestamped intake history

Visual Design System: Colors & Typography

Crafted around deep nocturnal navy surfaces, bioluminescent sapphire accents, and high-contrast geometric typography.

COLOR PALETTE & GLASSMORPHISM TOKENS

Navy Dark #080C16 Primary App Canvas	**Navy Deep** #0C1322 Elevated Card Surface	**Navy Light** #162035 Active Reminder Surface	**Sapphire Blue** #4A8CFF Primary Brand Accent
Azure Blue #2E6BFF Fluid Wave Depth	**Light Blue Glow** #82B1FF Specular Highlights	**Mint Cyan** #26E6E6 Bioluminescent Crest	**Glass Card Bg** #B315223D Frosted Overlay (70%)

TYPOGRAPHY SCALE & HIERARCHY (ROBOTO / GEOMETRIC SANS)

Alarm Hero Display	38sp • Black (-1.2sp tracking)	**Drink Water**
Clock Time Readout	32sp • ExtraBold (0.5sp tracking)	**03:30 PM**
Screen Title Header	26sp • ExtraBold (-0.5sp tracking)	**Hey, Vignesh / Hydration Stats**
Card Section Title	17sp–18sp • Bold	**6 / 8 Glasses Completed • 5 Day Streak!**
Body & Control Label	14sp • Bold / Medium	**Intake Reminders • Vibration Alert • 250 ml**
Caption & Chart Axis	11sp–12sp • Medium	**Target Line: 2.0L • Calculated automatically from last drink**



01. Home Dashboard

Real-Time 3D Fluid Tracking & Quick

1. Dynamic Greeting & Mute Shortcut

Displays personalized greeting, live date, and a 1-tap bell icon to toggle reminders.

2. 3D Animated Water Bottle Canvas

Dual sine-wave liquid layers, rising air bubbles, glass gloss & 0L–2.0L tick marks.

3. Glassmorphic Progress Summary

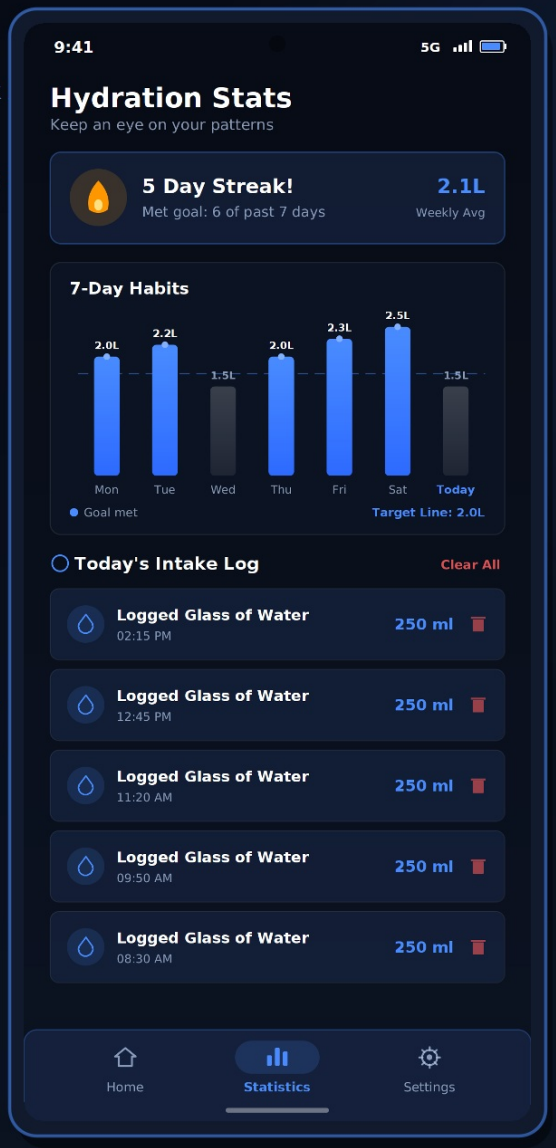
Shows completed glasses (6/8), liter total (1.5L/2.0L), and radial percentage ring (75%).

4. Google-Clock Reminder Card

Highlights the next scheduled drink alarm (03:30 PM) with interval label & switch.

5. Neon-Glow Drink Action FAB

One-tap +250ml water logging with instant mistake correction ('Subtract Last').



02. Hydration Stats

7-Day Habit Chart & Intake Log

1. Active Streak & Weekly Avg

Tracks consecutive goal-met days (5 Day Streak!) alongside 7-day average intake.

2. Custom 7-Day Canvas Bar Chart

Renders daily volume columns against a dashed target line (2.0L) with glowing caps.

3. Visual Goal Differentiation

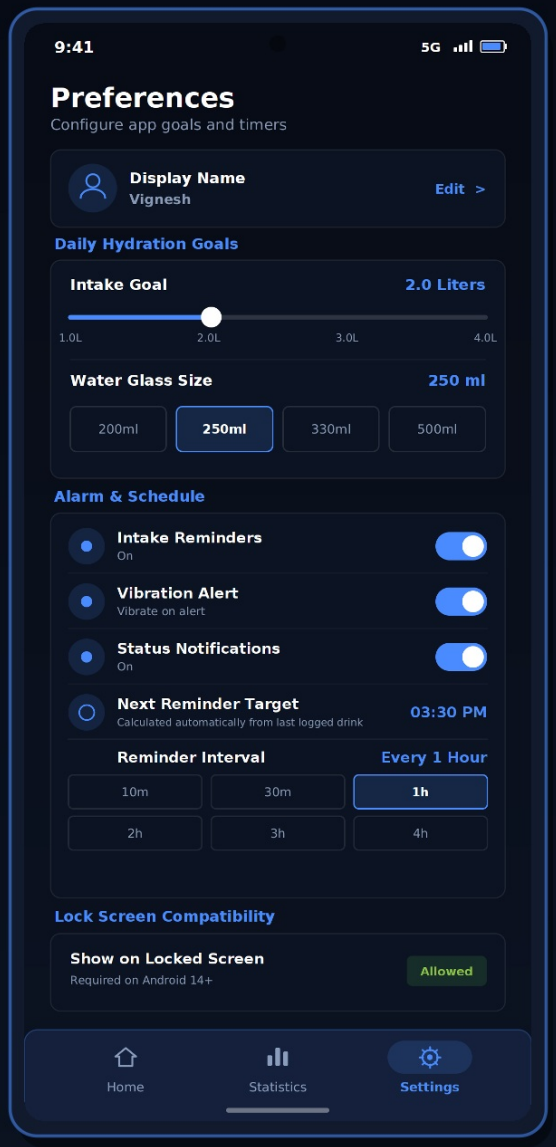
Days meeting target illuminate in Sapphire gradients; partial days remain muted.

4. Chronological Intake Log

Lists every logged glass with exact time (hh:mm a) and milliliter volume.

5. Granular Log Management

Allows deleting individual entries or clearing the entire day's log in 1 tap.



03. Smart Preferences

Goal Calibration & Granular Toggles

1. Personalized Display Name

Customizes morning and daily greetings across the Home dashboard.

2. Daily Goal & Glass Presets

Interactive 1.0L–4.0L slider paired with vessel chips (200ml, 250ml, 330ml, 500ml).

3. Granular Alarm & Haptic Switches

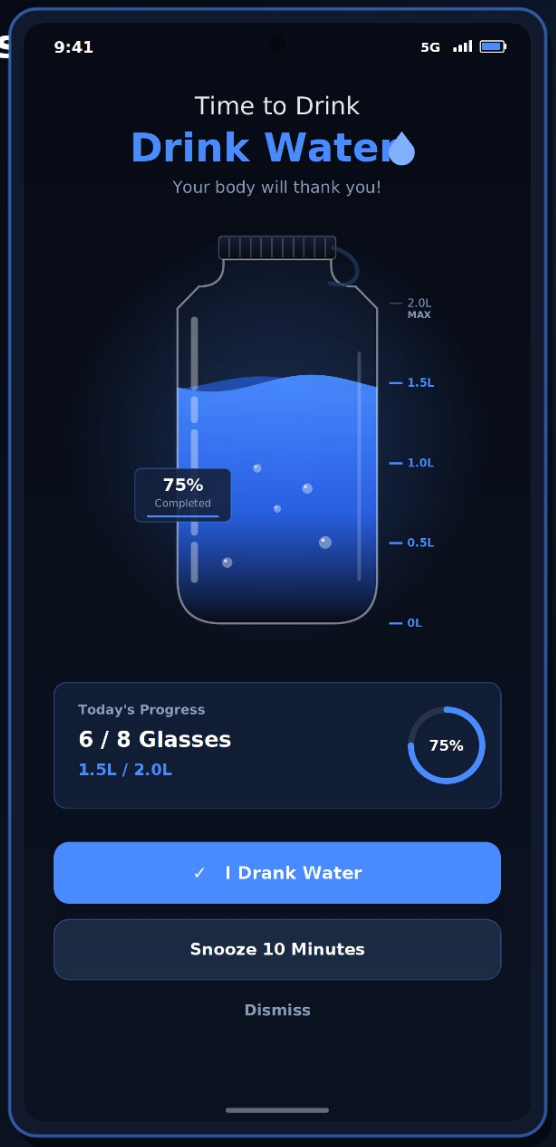
Independent toggles for Intake Reminders, Vibration Alert pulses & Status Notifications.

4. Flexible Interval & Custom Audio

6 interval presets (10m–4h) plus system ringtone or custom local audio file picker.

5. Android 14+ Lock-Screen Status

Live permission verification badges for Full-Screen Intent & Overlay permissions.



04. Lock-Screen Alert

Full-Screen Wake-Up Alarm UI

1. Lock-Screen Wake-Up Overlay

Launches over the device lock screen with turnScreenOn and showWhenLocked flags.

2. Looping Audio & Waveform Haptics

Plays selected alarm tone and repeating vibration waveform (420ms pulses).

3. Live 3D Bottle Context

Displays current daily hydration fill level right on the alarm screen for motivation.

4. One-Tap '✓ I Drank Water' CTA

Logs +250ml to Room DB, schedules next interval alarm, stops audio, and closes.

5. 10-Minute Snooze & Dismiss

Allows postponing the reminder by 10 minutes with a single tap when busy.

Technical Architecture & Android Engineering

Built with Clean MVVM Architecture, Reactive Kotlin StateFlows, Room Persistence, and Exact Alarm Scheduling.

UI & GRAPHICS LAYER (JETPACK COMPOSE)

- Custom WaterBottleCanvas with dual-phase sine wave paths & clipPath
- Custom 7-Day Canvas Bar Chart with nativeCanvas typography
- Edge-to-Edge WindowInsets & frosted NavigationBar with Crossfade
- Dedicated ReminderAlertActivity for lock-screen wake-up presentation

STATE & DOMAIN LAYER (MVVM VIEWMODEL)

- HydrationViewModel managing reactive StateFlows (progress, streak, logs)
- Automatic 7-day historical aggregation & streak calculation engine
- Dynamic next-reminder timestamp computation on every water log
- Persistable URI permission handling for custom local alarm audio

DATA & PERSISTENCE LAYER (ROOM SQLITE v4)

- HydrationLog Entity: id, amountMl, timestamp (chronological history)
- UserSettings Entity: userName, dailyGoalMl, glassSizeMl, intervalMinutes, reminderSoundUri, isReminderEnabled, isVibrationEnabled, isNotificationEnabled
- Coroutine Flow DAO queries for zero-latency UI updates

BACKGROUND ALARM PIPELINE (ALARMMANAGER)

- AlarmScheduler using setExactAndAllowWhileIdle for doze-mode reliability
- WaterReminderReceiver BroadcastReceiver with WakeLock acquisition
- Dual-path alert launch: Direct Activity start + FullScreenIntent Notification
- BootReceiver to automatically restore scheduled hydration alarms on reboot